HR Analytics

Employee Resignation Rate Prediction

Team Members :

ET24BTCO127	Mariyah Tajani
ET24BTCO130	Tasmi Hirpara
ET24BTCO142	Arshi Trivedi
ET24BTCO150	Honey Gajjar
ET24BTCO165	Khushi Patel

Guided By :

Prof. (Dr.) Dipali Kasat
Prof. (Dr.) Rachna Oza
Prof. Snehal Gandhi



INTRODUCTION

WHAT IS EMPLOYEE ATTRITION?

Employee attrition occurs when employees leave an organization voluntarily or otherwise.

- Increases recruitment and training costs
- Affects productivity and team stability
- Creates a need for effective employee retention strategies

OBJECTIVES

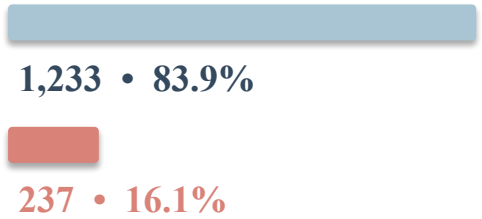
- Predict whether an employee will stay or leave.
- Identify factors influencing employee attrition.
- Apply KNN and ID3 Decision Tree algorithms.
- Compare model performance and support employee retention.

DATASET

IBM HR Analytics Employee Attrition dataset

1,470	35	0	0
EMPLOYEE RECORDS	ORIGINAL COLUMNS	MISSING VALUES	DUPLICATE ROWS

ATTRITION DISTRIBUTION



FEATURE GROUPS

- Personal and demographic
- Job profile and compensation
- Work Experience
- Satisfaction and work-life balance

DATA PRE-PROCESSING

01 Load & inspect › 02 Remove 4 columns › 03 Encode categories › 04 80:20 split › 05 Min-Max scaling

```
Dataset shape: (1470, 35)

Missing values: 0
Duplicate rows: 0

Attribution count:
Attribution
No     1233
Yes     237
Name: count, dtype: int64

Shape after removing unnecessary columns: (1470, 31)
Total features: 51

Training records: 1176
Testing records: 294

Final training shape: (1176, 51)
Final testing shape: (294, 51)
```

```
First five records:
   Age Attrition     BusinessTravel  DailyRate  ... YearsAtCompany
0   41       Yes      Travel_Rarely       1102  ...              6
1   49        No  Travel_Frequently        279  ...             10
2   37       Yes      Travel_Rarely       1373  ...              0
3   33        No  Travel_Frequently       1392  ...              8
4   27        No      Travel_Rarely        591  ...              2

[5 rows x 35 columns]
```

MODEL INPUT & OUTPUT

Input (X):	**51 encoded features**
Target (y):	**Attrition - Yes / No**

METHODOLOGY

Two classification algorithms are trained and evaluated on the same test records.

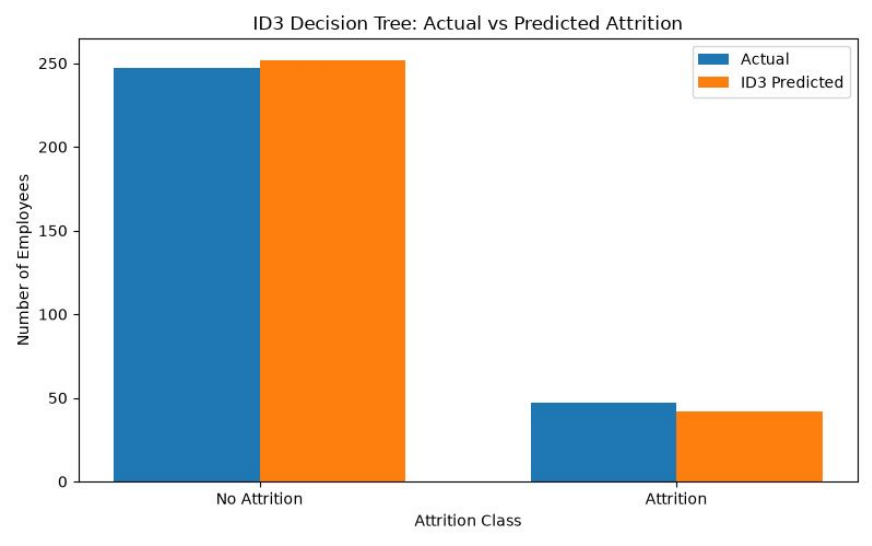
K-NEAREST NEIGHBORS (KNN)

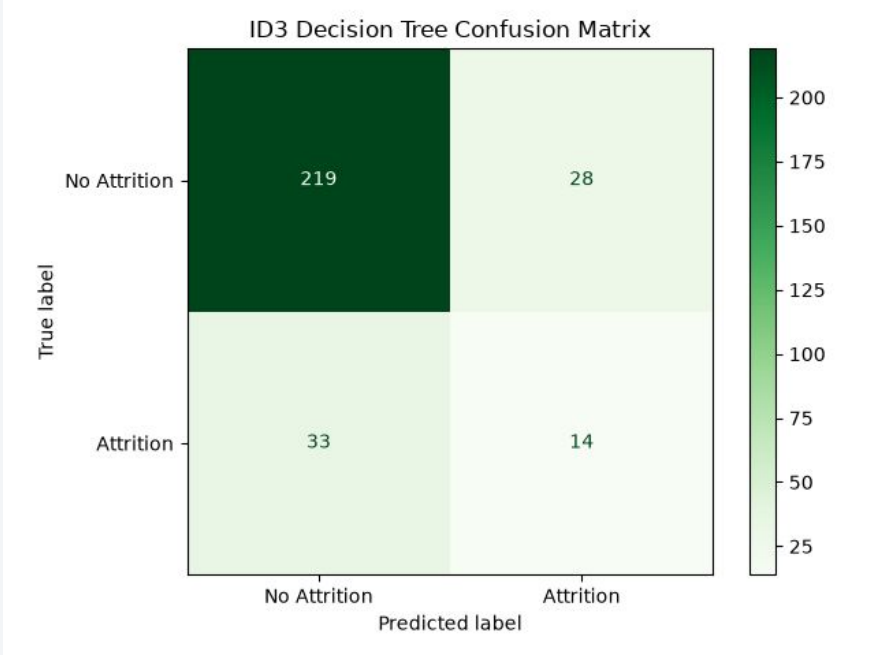
- Input: Uses 51 encoded employee features.
- Distance calculation: Measures the distance between the new employee record and training records.
- Neighbor selection: Identifies the five closest employee records.
- Majority voting: Counts the Attrition labels of the five neighbors.
- Prediction: Assigns the majority class — Yes (left) or No (stayed).

ID3 DECISION TREE

- Input: Uses employee features and known Attrition labels for training.
- Entropy: Measures uncertainty in the Attrition classes.
- Information Gain: Determines how effectively each feature separates employees who stayed and left.
- Feature Selection: Selects the feature with the highest information gain.
- Tree Construction: Repeats the splitting process to create branches and decision rules.
- Prediction: Follows the decision path to a leaf node and predicts Attrition.

DECISION TREE MODEL EVALUATION

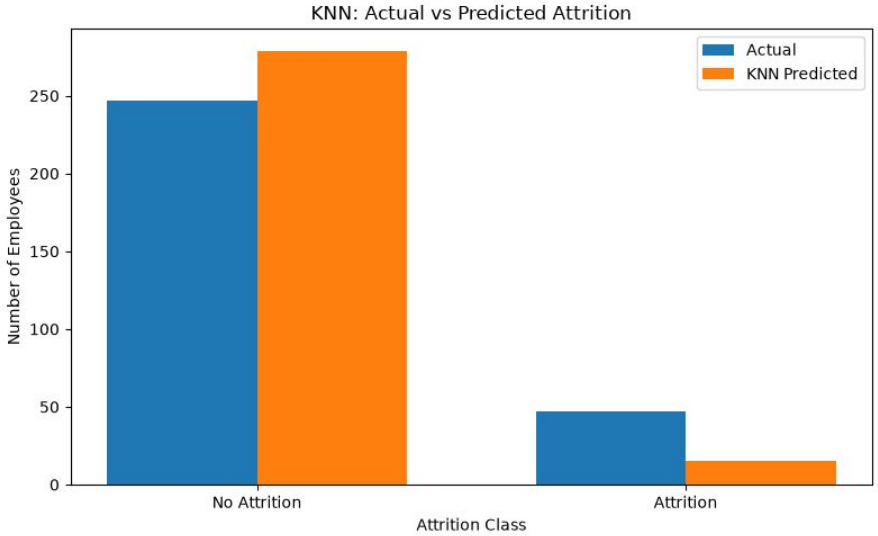


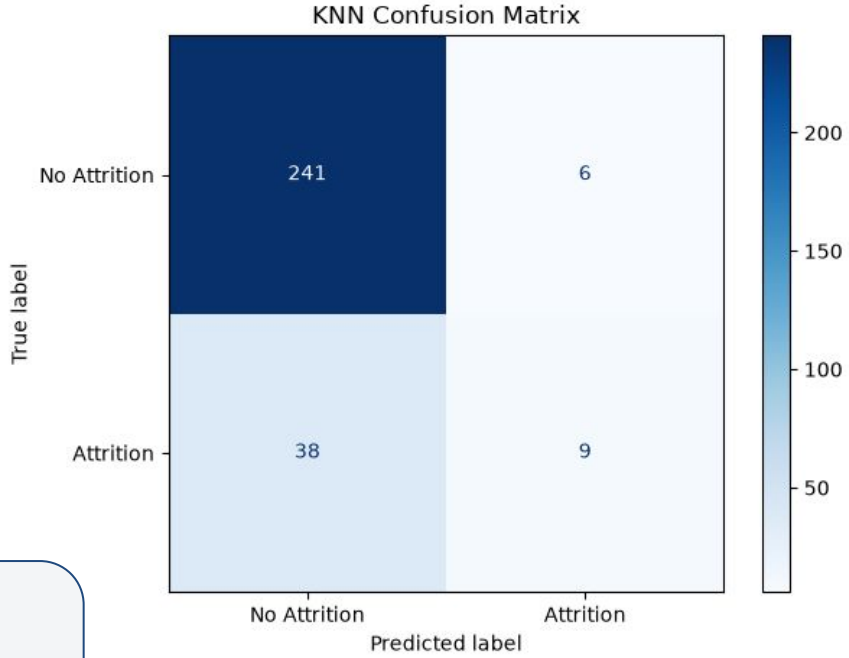


Key Finding

The ID3 model correctly classifies most employees who stay but misses many actual resignations, achieving only 29.79% recall.

KNN MODEL EVALUATION





Key Finding

The KNN model achieves high accuracy but tends to overpredict No Attrition, resulting in more missed resignations and a recall of just 19.15%.

PREDICTION RESULTS

ID3 DECISION TREE PREDICTION TABLE (First 10 Records)

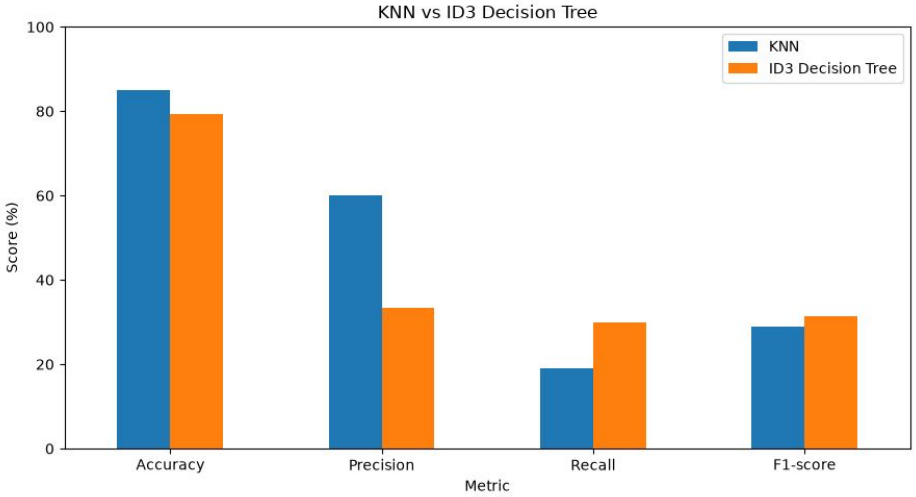
Actual Attrition	Predicted Attrition	Result
No	No	Correct
No	No	Correct
No	No	Correct
No	No	Correct
Yes	No	Incorrect
No	No	Correct
No	No	Correct
No	No	Correct
No	No	Correct
No	Yes	Incorrect

KNN PREDICTION TABLE (First 10 Records)

Actual Attrition	Predicted Attrition	Result
No	No	Correct
No	No	Correct
No	No	Correct
No	No	Correct
Yes	No	Incorrect
No	No	Correct
No	No	Correct
No	No	Correct
No	No	Correct
No	No	Correct

Both models predict attrition on unseen test records, comparing predicted labels with actual outcomes to identify correct and incorrect classifications

MODEL COMPARISON



MODEL COMPARISON

Metric	KNN	ID3 Decision Tree
Accuracy	85.03	79.25
Precision	60.00	33.33
Recall	19.15	29.79
F1-score	29.03	31.46

- **Learning Strategy:** KNN uses instance-based learning; ID3 builds a decision tree.
- **Decision Mechanism:** KNN uses distance and majority voting; ID3 uses entropy and information gain.
- **Attrition Detection:** KNN has higher precision (60.00%); ID3 has higher recall (29.79%).
- **Performance Trade-off:** KNN achieves higher accuracy; ID3 has a higher F1-score.
- **Key Limitation:** Both models miss many actual resignations.

LIMITATIONS & FUTURE IMPROVEMENTS

Limitations

- **KNN Parameter Selection:** The chosen value (k=5) may not be optimal.
- **Possible Overfitting:** The ID3 tree may overfit the training data if it grows too deep.
- **Class Imbalance:** Fewer resignation cases may reduce the model's ability to detect attrition.
- **Feature Sensitivity:** Irrelevant features may affect KNN distance calculations and ID3 splits.

Future Improvements

- Alternative Algorithms: Implement CART or Random Forest for comparison.
- Tree Pruning: Reduce unnecessary branches in ID3 to improve generalization.
- Class Balancing: Apply oversampling or undersampling to address class imbalance.
- Feature Selection: Identify and retain the most relevant attributes for prediction.

CONCLUSION

- Applied data preprocessing, categorical encoding, Min-Max normalization and classification to the HR Analytics dataset.
- Developed KNN and ID3 models to classify employees into Attrition (Yes/No).
- KNN achieved 85.03% accuracy, while ID3 achieved a higher recall of 29.79% and F1-score of 31.46%.
- Confusion matrix analysis revealed that both models struggled to identify employees who resigned.
- The study highlights the importance of class distribution and multiple evaluation metrics in assessing classification performance.

REFERENCES

[1] IBM HR Analytics Employee Attrition & Performance dataset (WA_Fn-UseC_-HR-EmployeeAttrition.csv). https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset

[2] F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011. https://scikit-learn.org/stable/modules/generated/sklearn.neighbors.KNeighborsClassifier.html

[3] T. Cover and P. Hart, “Nearest neighbor pattern classification,” IEEE Transactions on Information Theory, vol. 13, no. 1, pp. 21–27, 1967.

[4] J. R. Quinlan, “Induction of decision trees,” Machine Learning, vol. 1, no. 1, pp. 81–106, 1986.

[5] J. Han, M. Kamber and J. Pei, Data Mining: Concepts and Techniques, 3rd ed., Morgan Kaufmann, 2012.

Thank You

